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EDUCATION

PhD Epidemiology , <i>University of North Carolina at Chapel Hill</i>	2013
MSPH Epidemiology , <i>University of North Carolina at Chapel Hill</i>	2010
BS International Affairs , <i>Georgia Institute of Technology</i>	2007

PROFESSIONAL EXPERIENCE

University of North Carolina at Chapel Hill

Professor , Department of Epidemiology	2026 – present
Associate Professor , Department of Epidemiology	2022 – 2026
Faculty Fellow , Carolina Population Center	2023 – present
Assistant Professor , Department of Epidemiology	2020 – 2022
Research Assistant Professor , Department of Epidemiology	2015 – 2020
Postdoctoral Research Associate , Department of Epidemiology	2013 – 2015

HONORS

Joan Heckler Gillings Endowed Professorship <i>University of North Carolina at Chapel Hill</i>	2025 – 2028
Faculty Award for Excellence in Graduate Student and Academic Program Support <i>University of North Carolina at Chapel Hill</i>	2026
Editor's Choice Paper <i>Epidemiology</i>	2020
Reviewer of the year <i>American Journal of Epidemiology</i>	2015
Berton H. Kaplan Student Publication Award <i>University of North Carolina at Chapel Hill</i>	2013
Sidney Kark Award for Distinguished Teaching Assistant <i>University of North Carolina at Chapel Hill</i>	2013
Delta Omega Public Health Honor Society, Theta Chapter	2013
Society for Epidemiologic Research Poster Award	2012
Graduate Merit Assistantship <i>University of North Carolina at Chapel Hill</i>	2008–2009

MEMBERSHIPS

1. American Statistical Association, 2019 – present
2. Society for Epidemiologic Research, 2009 – present
3. American Association for the Advancement of Science, 2007 – 2010

BIBLIOGRAPHY AND PRODUCTS OF SCHOLARSHIP

BOOKS AND CHAPTERS

1. Keil A, **Edwards JK**. Bias in environmental epidemiology. In: Encyclopedia of Environmental Health, 2e. 2019: 288-300

REFEREED PAPERS

1. Zivich PN, **Edwards JK**, Lucas VA, Shook-Sa BE, Brookhart MA, Cole SR. Computing Assumption-Lean Bounds for Per-Protocol Effects. *Trials* 2026; *In Press*
2. Zivich PN, Cole SR, Greifer N, Montoya LM, Kosorok MR, **Edwards JK**. Confidence Regions for Multiple Outcomes, Effect Modifiers, and Other Multiple Comparisons. *American Journal of Epidemiology* 2026; *In Press*
3. Neuroth LM, Martinez M, **Edwards JK**, Zivich PN. Pooled multinomial logistic regression for parametric g-computation in the presence of competing events. *Epidemiology* 2026; *In Press*
4. Carey-Ewend K, Shepard H, Yang G, Odas M, Peter EE, Dominick MC, Rutha I, Amagai K, Vasquez-Martinez B, Mulei LA, Chhetri S, Kharabora S, Adams JK, Lin F, Bailey J, **Edwards JK**, Gower EW, Juliano JJ, Ngasala BE, Lin JT. Frequent Plasmodium ovale recurrence in coastal Tanzania, *Clinical Infectious Diseases*, 2026; *In Press*
5. Cole SR, Zivich PN, Shook-Sa BE, Richardson DB, Hudgens MG, **Edwards JK**. Combining observational studies to reduce multiple biases. *Epidemiology* 2026; *In press*
6. **Edwards JK**, Zivich PN, Shook-Sa BE, Cole SR. The Rogan-Gladen estimator for outcome misclassification. *American Journal of Epidemiology*. 2026; *In Press*
7. François-Zafka R, Westreich D, Banek K, Kashamuka MM, White SJ, Bala JA, Nkalani M, Kihuma G, Atibu J, Mahilu GE, Thwai KL, Kokolomami J, Emch M, **Edwards JK**, Juliano JJ, Tshefu A, Parr JB. Challenges in malaria diagnosis and treatment in Kinshasa Province, Democratic Republic of the Congo. *Clinical Infectious Diseases*. 2026; *In Press*
8. Momkus J, Harris KM, **Edwards JK**, Yang YC, Martin CL, Aiello AE. A Pathogen Penalty? Associations between Persistent Infections and Biological Aging in the US. *Journal of Infectious Diseases*. 2026; *In Press*
9. Latour CD, **Edwards JK**, Jonsson Funk M, Suarez E, Boggess K, Wood ME. Bias in studies of prenatal exposures using real-world data due to pregnancy identification method. *American Journal of Epidemiology*. 2026; *In Press*
10. Abuogi LL, Carlucci JG, Edmonds A, Regula L, Sugandhi N, Yin DE, **Edwards JK**. A systematic review of the use of antiretroviral drugs to prevent HIV in neonates and infants. *Journal of the Pediatric Infectious Diseases Society*. 2026 Jul 2;piago51.

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12. Ingol TT, **Edwards JK**, Wood ME, Martin CL, Dichter GS, Feldman K, Kinlaw AC, Daniels JL. Labor epidural analgesia and autism spectrum disorder in offspring: A population-based case-control study. *Autism Research*. 2026:e70210.
13. Wood ME and **Edwards JK**. Conceiving of Study Designs to Evaluate Preconception Treatment Decisions. *JAMA Network Open*. 2026; 9(3): e260664
14. Carey-Ewend K, Marten A, Muller J, Peter EE, Odas M, Dominick MC, Muller M, Chhetri S, Amagai K, Rutha I, Kisandu F, Beka L, Kharabora O, Ropkin-Hall ZR, Bailey J, **Edwards JK**, Gower EW, Juliano JJ, Ngasala BE, Lin JT. Seasonal variation and interspecies dynamics among *Plasmodium falciparum* and ovale species in Bagamoyo, Tanzania. *Journal of Infectious Diseases*. 2026; 233(2):391-400.
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19. Zivich PN, Klose M, DeMonte JB, Shook-Sa BE, Cole SR, **Edwards JK**. An Improved Pooled Logistic Regression Implementation. *Epidemiology*. 2025; 37(3): 311-314
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23. Ross RK, Fox MP, Lesko CR, Rudolph JE, Zalla LC, **Edwards JK**. Using measurement error parameters from validation data. *Epidemiology*. 2025; 37(1):67-72
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- teritis risk in a Nicaraguan birth cohort. *The American Journal of Clinical Nutrition*. 2025; 122(2):488-501
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69. *Zalla LC, **Edwards JK**, Rudolph JE, Mulholland GE, Cole SR. Data movies: a tool for public health. *Epidemiology*. 2023; 34(6):854-5
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73. Ross RK, Cole SR, **Edwards JK**, Westreich D, Daniels J, Stringer JSA. Accounting for Nonmonotone Missing Data using Inverse Probability Weighting. *Statistics in Medicine*. 2023; 42 (23), 4282-4298
74. Keil AP, Zadrozny S, **Edwards JK**. A review and synthesis of multi-level models for causal inference with individual level exposures. *Current Epidemiology Reports*. 2023; Aug 10:1-9.
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 186. Rudolph JE, Cole SR, **Edwards JK**. Parametric assumptions equate to hidden observations: comparing the efficiency of nonparametric and parametric models for estimating time to AIDS or death in a cohort of HIV-positive women. *BMC Medical Research Methodology*. 2018; 18:142
 187. **Edwards JK**, Cole SR, Moore RD, Mathews WC, Kitahata M, Eron JJ. Sensitivity analyses for misclassification of cause of death in the parametric g-formula. *American Journal of*

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 191. Keil AP and **Edwards JK**. You are smarter than you think: (super) machine learning in context. *European Journal of Epidemiology*. 2018; 33(5): 437-440
 192. Bengtson AM, Pence BW, Eaton EF, **Edwards JK**, Eron JJ, Mathews WC, Mollan K, Moore RD, O'Cleirigh C, Geng E, Mugavero MJ. Patterns of Efavirenz Use as First Line Antiretroviral Therapy in the United States: 1999-2015. *Antiviral Therapy*. 2018; 23:363-372
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 195. Rudolph JE, Cole SR, **Edwards JK**, Moore R, O'Cleirigh C, Mathews WC, Christopoulos K. At-risk alcohol use among HIV-positive patients and completion of patient-reported outcomes. *AIDS and Behavior*. 2018; 22(4): 1313-1322
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- schooling on age-disparate relationships and number of sexual partners among young women in rural South Africa enrolled in HPTN 068. *AIDS*. 2017; 76(5): e107-e114
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 224. **Edwards JK**, Cole SR, Westreich D. All your data are always missing: incorporating bias due to measurement error into the potential outcomes framework. *International Journal of Epidemiology*. 2015; 44(4), 1452-1459.
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235. **Edwards JK**, Cole SR, Troester MA, Richardson DB. Accounting for Misclassified Outcomes in Binary Regression Models Using Multiple Imputation With Internal Validation Data. *American Journal of Epidemiology*. 2013;177(9):904-912
236. Weir SS, Li J, **Edwards JK**, Gandhi AD, Yingying H, Suchindran CM, Chen X. Exploring Venue-Associated Risk: A Comparison of Multiple Partnerships and Syphilis Infection Among Women Working at Entertainment and Service Venues. *AIDS and Behavior*. 2013; 18(2), 153-160
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239. Denslow SA, **Edwards JK**, Horney J, Peñas R, Morgan D. The effect of water and sanitation infrastructure on the prevalence of diarrheal disease in rural Nicaragua. *BMC International Health and Human Rights* 2010; 10(1), 30

OTHER PRODUCTS OF SCHOLARSHIP

1. Cole SR, **Edwards JK**, Breskin A, Rosin S, Zivich PN, Shook-Sa BE, Hudgens MG. Cole et al. Respond to “Combining Information From Diverse Sources”. *American journal of epidemiology*. 2024 May;193(5):751-2.
2. Wirth KE, **Edwards JK**, Feinstein L, Breskin A. When emulating a trial, do as the trialists do: Missteps in estimating relative effectiveness of a severe acute respiratory syndrome coronavirus 2 vaccine booster dose. *Clinical Infectious Diseases*. 2023 Jan 1;76(1):176-7.
3. Lesko CR, Fox MP, **Edwards JK**. Epidemiologic questions require context. *American Journal of Epidemiology*. 2023; 192(4):683
4. Lesko CR, Fox MP, **Edwards JK**. Describing health threats. *American Journal of Epidemiology*. 2022; 191(12): 2073-2074
5. Lesko CR, Fox MP, **Edwards JK**. Lesko et al. Respond to “The Importance of Descriptive Epidemiology”. *American journal of epidemiology*. 2022 Nov 19;191(12):2073-4
6. **Edwards JK**, Cole SR, Adimora AA. Remdesivir and COVID-19. *The Lancet*. 2020 Oct 3;396(10256):953.
7. Westreich D, van Smeden M, **Edwards JK**. COMMENT ON WILLIAMSON ET AL.(OpenSAFELY): the Table 2 fallacy in a study of COVID-19 mortality risk factors. *Epidemiology*. 2021; 32(1):e1-2
8. **Edwards JK**, and Lessler J. What now? Epidemiology in the wake of a pandemic. *American Journal of Epidemiology*. 2020; 190(1):17-20
9. Lesko CR, Buchanan, AL, Westreich D, **Edwards JK**, Hudgens MG, Cole SR. Re: Generalizing

- study results: a potential outcomes perspective. The authors respond. *Epidemiology*. 2018; 29(2), e14-e15
10. Cole SR, Chu H, Brookhart MA, **Edwards JK**. Re: Dogmatists cannot learn. The authors respond. *Epidemiology*. 2017; 28(6), e62-e63

INVITED PRESENTATIONS

1. Improving decisions in the void: combining statistical and mechanistic models to improve early outbreak response. Society for Epidemiologic Research Annual Meeting, Boston, MA. June 2025
2. Of pregnancy and target trials. Society for Epidemiologic Research Annual Meeting, Boston, MA. June 2025
3. Improving estimates of mortality risk functions in settings with incomplete data linkage. International Workshop on HIV Observational Databases. Toledo, Spain. March 2025
4. Semiparametric g-computation to account for baseline confounding in observational studies with survival endpoints. World Congress of Epidemiology, Cape Town, South Africa, 25 September 2024
5. Outcome measurement error in time-to-event analyses. Victorian Centre for Biostatistics, Melbourne Australia, 22 May 2024
6. Questions and Answers about Integrase Inhibitors and Weight Gain. International Workshop on HIV Observational Databases. Vilamoura, Portugal. March 2024.
7. Outcome Measurement Error in Survival Functions. Karolinska Institute. Stockholm, Sweden, 24 October 2023
8. Measurement Error in Survival and Risk Functions. Boston University. Boston, MA, 5 September 2023
9. Merging statistical models, mechanistic models, and causal inference in outbreak response. Harvard School of Public Health. Boston, MA, 6 April 2023
10. Imputation approaches to account for measurement error in marginal structural models. Center for Outcomes Research. McGill University, Montreal Canada, 22 February 2023
11. Accounting for outcome misclassification in causal risk functions. Joint Statistical Meetings. Washington DC, 9 August 2022
12. Fusion designs for measurement error. Society for Epidemiologic Research. Chicago IL, 16 June 2022
13. Exploring the elevation in mortality among people with HIV in the era of effective therapy. Epidemiology Student Association Seminar Series. University of California – Los Angeles, Department of Epidemiology, 19 October 2021
14. Fractional Imputation. Society for Epidemiologic Research. Online. 25 Jun 2021
15. Estimating the sizes of key populations: considerations for design and analysis. CDC Key Population Surveillance and Estimates: Recent Advances and Future Directions Technical Working Group. 24 May 2021
16. Exploring the elevation in mortality among people with HIV and interventions for mitigation. University of California - Davis, Department of Epidemiology. 10 May 2021
17. Exploring the elevation in mortality among people with HIV. Harvard TH Chan School of Public Health Department of Biostatistics: HIV Working Group. 7 May 2021
18. Knowledge synthesis: informing decisions with incomplete data. University of Pennsylvania Seminar in Advanced Epidemiologic Methods. 23 April 2021
19. Exploring the elevation in mortality among people with HIV. University of Pennsylvania Center for Clinical Epidemiology and Biostatistics Departmental Seminar. 8 April 2021

20. Zen and the art of asking good questions, Society for Epidemiologic Research Annual Meeting. Online. 16 December 2020
21. Being principled about when we can be pragmatic: mechanics behind new user designs, Society for Epidemiologic Research and International Society for Pharmacoepidemiology joint webinar. 29 January 2020
22. A flexible approach to account for exposure measurement error in counterfactual risk functions, Joint Statistical Meetings, Denver, CO, 31 July 2019
23. Competing events: practical illustrations, Society for Epidemiologic Research Annual Meeting, Seattle, WA, 19 June 2019
24. Missing data, Society for Epidemiologic Research Annual Meeting, Seattle, WA, 21 June 2019
25. Estimating policy relevant causal effects with inverse probability weights, Keynote address at the SERdigital symposium, 13 March 2019
26. A hitchhikers guide to causal inference, Causal Inference Research Group, University of North Carolina, Chapel Hill, 7 September 2018
27. Causal inference across space and time, Society for Epidemiologic Research Annual Meeting, Baltimore, MD, 23 June 2018
28. An imputation approach to account for measurement error in marginal structural models, International Biometric Society Eastern North American Regional Meeting, Atlanta, GA, 27 March 2018
29. Integrating data from key population surveys into surveillance efforts, Measurement and Surveillance of HIV Epidemics Scientific Symposium, Johannesburg, South Africa, 7 December 2017
30. Estimating sizes of key populations as a missing data problem, UNAIDS Reference Group Meeting, London, United Kingdom, 19 October 2017
31. A randomized trial of data adaptive doubly robust estimators versus standard practice, Society for Epidemiologic Research Annual Meeting, Seattle, WA, 22 June 2017
32. A method to compare viral suppression over time in observational studies with competing events: Applied example to estimate the comparative effectiveness of raltegravir vs efavirenz, International Workshop on HIV Observational Databases, Lisbon, Portugal, 31 March 2017
33. A geographic approach to extrapolating characteristics of key populations, Population Size Estimation Technical Consultation, CDC, Atlanta, GA, 31 January 2017
34. Estimating the size, location and key characteristics of most-at-risk populations, Measurement and Surveillance of HIV Epidemics Scientific Symposium, Tallinn, Estonia, 27 October 2016
35. Disparities in the HIV care continuum in the Dominican Republic, Social Epidemiology Seminar, UNC, Chapel Hill, NC, 12 October 2016
36. Extrapolation of data from key population surveys and programs, AIDS, Durban, South Africa, 18 July 2016
37. Scaling up interventions on the HIV care continuum: a practical example of meeting in the middle, Epidemiology Congress of the Americas, Miami, FL, 24 June 2016
38. External validity and the transportability of internally valid effects, Epidemiology Congress of the Americas, Miami, FL, 22 June 2016
39. Measurement error and causal inference: Incorporating measurement error into the potential outcomes framework, Workshop on Measurement Error and Complex Data, College Station, TX, 22 April 2016
40. Estimating risk functions for cause-specific mortality when cause of death may be misclassified, International Workshop on HIV Observational Databases, Budapest, Hungary, 8 April 2016

41. Quantitative methods to extrapolate size estimates: a geographic approach, Strategic information for key populations technical consultation, UNAIDS, Geneva, Switzerland, 9 March 2016
42. Benefits of unique identifiers for studying the HIV care and treatment cascade, Strategic information for key populations technical consultation, UNAIDS, Geneva, Switzerland, 8 March 2016
43. Age at entry into care, timing of antiretroviral therapy initiation, and 10-year mortality among HIV-seropositive adults in the United States, International Workshop on HIV Observational Databases, Catania, Sicily, 27 March 2015
44. Missing data in epidemiologic analyses, UNC Annual Epidemiologic Methods Workshop, 10 October 2014
45. Estimating effects of interventions, Society for Epidemiologic Research, Seattle, WA, 27 June 2014
46. Loss to follow-up and mortality among US HIV-infected antiretroviral therapy initiators, Society for Epidemiologic Research, Seattle, WA, 26 June 2014
47. Missing data and causal inference, Causal inference research group, University of North Carolina, Chapel Hill, 7 February 2014
48. Incorporating retention in care into estimates from observational HIV cohort studies: implications for evaluation and inference, Biostatistics seminar, University of California, Berkeley, School of Public Health, 13 November 2013
49. Methods to account for misclassification of cause-specific mortality, Society for Epidemiologic Research, Boston, MA, 21 June 2013
50. Comparison of three causal models to control time-varying confounding in a cohort of bone marrow transplant recipients, Causal inference research group, University of North Carolina, Chapel Hill, 7 September 2012

TEACHING

UNIVERSITY OF NORTH CAROLINA AT CHAPEL HILL

Gillings School of Global Public Health

Lead Instructor

EPID 717: Advanced Epidemiologic Methods 2024 - present

EPID 722: Advanced Epidemiologic Methods 2021 - 2024

EPID 886: Independent Study in Epidemiology

1. Competing events and cumulative counts (1 student) 2025
2. Transportability and competing events (1 student) 2024
3. Semiparametrics and measurement error (1 student) 2022

EPID 722: Epidemiologic Analysis of Time to Event Data 2018, 2019

EPID 700: SAS Programming and Data Management 2012

Coinstructor

EPID 722: Epidemiologic Analysis of Time to Event Data 2020

Guest Lecturer

EPID 722: Epidemiologic Analysis of Time to Event Data 2013, 2015 - 2017

EPID 718: Analytic Methods in Observational Epidemiology	2014
EPID 710: Fundamentals of Epidemiology	2015
EPID 733: Clinical Trials	2018
EPID 804: Design of Clinical Research Studies	2022 - 2026
EPID 715: Theory and Quantitative Methods in Epidemiology	2025, 2026

Teaching Assistant

EPID 722: Epidemiologic Analysis of Time to Event Data	2012, 2013
EPID 715: Theory and Quantitative Methods in Epidemiology	2011
EPID 710: Fundamentals of Epidemiology	2009

School of Law

Guest Lecturer

Scientific Evidence	2021
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OTHER SHORT COURSES, SEMINARS, AND WORKSHOPS

Instructor , Introduction to Causal Inference and the G-methods International Summer School of Epidemiology, Ulm University	2025
Instructor , Principles of Epidemiology International Summer School of Epidemiology, Ulm University	2024
Organizer , Course Development and Revision Workshop Society for Epidemiologic Research Annual Meeting	2022 - 2024
Guest lecturer , Advanced Topics in Epidemiologic Methods University of Pennsylvania, Philadelphia, PA	2021 - 2023
Co-instructor , Capture-Recapture Workshop North Carolina State Health Department, Raleigh, NC	2019
Instructor , Priorities for local AIDS control efforts workshop Cape Town, South Africa	2016

MENTORING

STUDENT MENTORING, CURRENT

Doctoral thesis committee chair (in progress): 1

Doctoral thesis committees (in progress): 11

Current doctoral students (pre-dissertation phase): 2

STUDENT MENTORING, COMPLETED

Doctoral thesis committee chair (completed): 7

1. Rachel Sendor, Epidemiology, UNC, 2026: "Dengue and chikungunya virus epidemiology and vaccine considerations in the Democratic Republic of the Congo"
2. Ning Zhang, Epidemiology, UNC, 2026: "Use of Nirmatrelvir-Ritonavir in Acute SARS-CoV-2 Infection and Implications for Long COVID Prevention"

3. Caitlin Cassidy, Epidemiology, UNC, 2026: "False-positive malaria rapid diagnostic tests and antimalarial overuse among children under 5 in Uganda"
4. Kelsey Landrum, Epidemiology, UNC, 2024: "The children who go uncounted: quantifying the global surgical disease burden of major congenital gastrointestinal anomalies"
5. Leah Sadinski, Epidemiology, UNC, 2024: "Blood pressure trajectories and mortality among people with and without HIV in a United States-based cohort study"
6. Lauren Zalla, Epidemiology, UNC, 2022: "Racial disparities in HIV care and mortality in the modern treatment era"
7. Grace Mulholland, Epidemiology, UNC, 2021: "Using enriched clinical cohort data to inform interventions for TB control in the Lake Victoria Region of East Africa"

MSPH thesis committees (completed): 2

1. Grace Mulholland, Epidemiology, UNC, 2016
2. Jackie Rudolph, Epidemiology, UNC, 2016

Doctoral thesis committees (completed): 50

1. Carson Telford, Epidemiology, UNC, 2026
2. Sherin Ismail Atta, Epidemiology, UNC, 2026
3. Monica Martinez, Epidemiology, UNC, 2026
4. Ruthly Francois-Zafka, Epidemiology, UNC, 2025
5. Ting Chen, Maternal and Child Health, UNC, 2025
6. Tim Feeney, Epidemiology, UNC, 2025
7. Yen Chang, Biostatistics, UNC, 2025
8. Luis Zambrana, Epidemiology, UNC, 2025
9. Kelly Carey-Ewend, Epidemiology, UNC, 2025
10. Jennifer Momkus, Epidemiology, UNC, 2025
11. Chase Latour, Epidemiology, UNC, 2025
12. Taniqua Ingol, Epidemiology, UNC, 2025
13. Brooke Bullington, Epidemiology, UNC, 2025
14. Catherine Li, Epidemiology, UNC, 2025
15. Kadija Tahlil, Epidemiology, UNC, 2024
16. Brittany Trottier, Epidemiology, UNC, 2024
17. Opal Patel, Epidemiology, UNC, 2024
18. Rebecca Rubenstein, Epidemiology, UNC, 2024
19. Rachael Ross, Epidemiology, UNC, 2023
20. Kate Lemasters, Epidemiology, UNC, 2023
21. Emilie Duchesneau, Epidemiology, UNC, 2023
22. Erica Zeno, Epidemiology, UNC, 2023
23. Samantha Tulenko, Epidemiology, UNC, 2023
24. Sam Rosin, Biostatistics, UNC, 2023
25. Camille Morgan, Epidemiology, UNC, 2023
26. Matthew Romo, Epidemiology, CUNY, 2022
27. Charley Gaber, Epidemiology, UNC, 2022
28. Sophie Mayer, Epidemiology, UNC, 2022
29. Arti Virkud, Epidemiology, UNC, 2022
30. Denise St. Jean, Epidemiology, UNC, 2022

31. Amrita Rao, Epidemiology, JHU, 2022
32. Hillary Topazian, Epidemiology, UNC, 2021
33. Kelsey Sumner, Epidemiology, UNC, 2021
34. Lindsey Filiatreau, Epidemiology, UNC, 2021
35. Nosipho Shangase, Epidemiology, UNC, 2021
36. Cedar Mitchell, Epidemiology, UNC, 2021
37. Bryan Blette, Biostatistics, UNC, 2021
38. Molly Deutsch-Feldman, Epidemiology, UNC, 2020
39. Nicholas Brazeau, Epidemiology, UNC, 2020
40. Tiffany Breger, Epidemiology, UNC, 2020
41. Phyo Htoo, Epidemiology, UNC, 2020
42. Nicole Frascino, Epidemiology, UNC, 2020
43. Rebecca Stebbins, Epidemiology, UNC, 2020
44. Jackie Rudolph, Epidemiology, UNC, 2019
45. Michael Webster-Clark, Epidemiology, UNC, 2019
46. Lisa Albert, Epidemiology, UNC, 2019
47. Marie-Josophe Horner, Epidemiology, UNC, 2018
48. Terra Fatuski, Epidemiology, UNC, 2018
49. Marie Stoner, Epidemiology, UNC, 2017
50. Bradley Saul, Biostatistics, UNC, 2017

Other student mentoring

EQUITY MPH Capstone adviser, 2022 - 2024

Curriculum Committee Member, MCH, 2022 - 2024

POSTDOC MENTORING: 3

1. Abagael Sykes (2024 - 2026)
2. Grace Mulholland (2021 - 2023)
3. Sabrina Zadrozny (2015 - 2017)

FACULTY MENTORING

NIH K Award Mentoring Committees: 3

1. Sara Levintow, UNC, K01: Investigating SARS-CoV-2 transmission dynamics in jails to address the COVID-19 public health emergency (2022 - 2025)
2. Paul Zivich, UNC, K01: Combining sources of information to improve HIV pre-exposure prophylaxis (2023 - 2028)
3. McKaylee Robertson, CUNY, K01: Examining the Interplay of COVID-19 and HIV: Impacts on HIV Care, Disparities, and Long-Term Outcomes (2024 - 2029)

Departmental Mentoring Committees: 2

FUNDING

CURRENT

NIH R01AI198037 (Edwards)

6/25/2026 - 5/31/2031

Answering important questions in HIV using mismeasured clinical endpoints

Routinely collected clinical data are valuable for HIV research but do not accurately capture all clinical outcomes, particularly chronic disease outcomes that are increasingly important as people with HIV age. The proposed work will develop new statistical approaches to improve HIV studies that rely on data with error-prone outcome measurements. These new methods will be used to address timely and important scientific questions in HIV related to heart disease, cancer, and mortality.

Role: PI

NIH R01AI157758 (renewal) (Edwards/Cole)

2/2/2026 - 1/31/2031

Improved analyses of experiments and observational studies in HIV

More accurate and robust health knowledge is a cornerstone of better medical and health policy in HIV and beyond. Results from both randomized trials and observational studies in HIV can be made more accurate and robust by combining multiple data sources and leveraging cutting-edge methods in epidemiology, statistics, and causal inference. In this work, we propose new study designs and estimators when leveraging enriched data in study subsets or combining data from multiple study sources.

Role: MPI

R01AG077947 NIH (Martin/Hargrove)

07/01/2022 - 03/31/2027

Multilevel Forms of Structural Racism and Racial Inequalities in ADRD Risk

The proposed study will create a database of multilevel, repeated, and prospective measures of structural racism; investigate the extent to which structural racism in education, residential, and criminal justice contexts shape Black-White inequalities in AD/ADRD risk; and use causal modeling techniques to evaluate the impact of structural interventions on documented racial inequalities in biological risk factors for AD/ADRD

Role: Co-investigator, 5% effort

P01HD112215 NIH (Hosseinipour)

04/01/2023 – 03/31/2028

Preventing Infant Infections with Implementation Science in Malawi

The Preventing infant infections by Implementation Science in Malawi (PRI3SM) project will conduct 3 related studies addressing the main reasons for pediatric HIV infections in Malawi, supported by administrative, implementation science and data science and analytical cores.

Role: Co-investigator, 5% effort

NU38FT000012 CDC (Lessler/Powers)

09/30/2023 – 09/29/2028

Atlantic Coast Center for Infectious Disease Dynamics and Analytics

To advance the state of the art in research and practice using modeling and analytics to control and

respond to infectious disease threats, with a focus on ensuring that analytic outputs are relevant to those setting public health policy across scales (i.e., from the national to the community level).
Role: Co-investigator, 10% effort

PAST

R61AI174285 NIH/NIAID (Rutstein/Muessig)

07/01/2022 - 06/30/2027

Meet me where I am: A multilevel strategy to increase PrEP uptake and persistence among young sexual and gender minority men in rural NC

This project partners with state and local health department sexually transmitted infection clinics to examine a multilevel intervention targeting PrEP use barriers at the policy, clinic, provider, and PrEP user levels, with the goal of increasing PrEP uptake and persistence among YSGM via an integrated, cost-effective, and scalable strategy.

Role: Co-investigator, 7% effort

NIH/NIAID R01AI157758 (Edwards/Cole)

8/1/2020 - 7/31/2025 \$3,609,505

Improved analyses of experiments and observational studies in HIV

In this work, we propose new and improved estimators of the risk (or survival) function to improve inference from randomized and nonrandomized HIV studies using new methods at the interface of statistics, epidemiology, causal inference, and artificial intelligence.

Role: MPI, 32% effort

U01HL146194 NIH (Flores-Moore)

01/10/2013 - 03/31/2026

University of North Carolina Womens Interagency HIV Study UNC WIHS

The UNC Center for AIDS Research provides essential infrastructure support for the wide range of HIV related research for its members. This research contributes to the major goals of Ending the HIV Epidemic and HIV prevention and cure, and has overarching public health impact.

Role: Co-investigator, 8% effort

WHO (Edwards)

3/1/2025 - 4/30/2025 \$12,000

Systematic review of infant postnatal prophylaxis for HIV

To synthesize existing research on the use of antiretroviral drugs to prevent vertical HIV transmission in neonates and infants. The outcome of this synthesis is intended to guide discussion on recommendations for postnatal infant prophylaxis against vertical HIV infection in WHO guideline development meetings as well as position results to be optimally disseminated to a range of international, national and subnational stakeholders.

Role: PI, 5% effort

NIH/NIGMS R01GM140564 (Edwards/Lessler)

2/1/2021 - 1/31/2025 (in NCE) \$1,533,472

Merging machine learning and mechanistic models to improve prediction and inference in emerging epidemics

The purpose of the proposed project is to improve inference, forecasting and decision making in response to emerging infectious diseases by developing a framework to integrate mechanistic and statistical approaches to epidemic modeling and causal inference. Approaches developed will be validated and applied prospectively to reduce morbidity and mortality in emerging public health

crises.

Role: MPI, 23% effort

NIH UG3HL156389 (Mutale/Herce)

09/10/2020 – 8/31/2024 \$262,423 (NCE)

Application of Implementation Science approaches to assess the effectiveness of Task-shifted WHO-PEN to address cardio metabolic complications in people living with HIV in Zambia

This study aims to apply empirically supported implementation science strategies, including task shifting, audit and feedback, and eHealth clinical decision support to integrate management of cardiovascular disease risk factors and cardiometabolic complications of HIV within a robust HIV service delivery platform in Lusaka, Zambia.

Role: Co-investigator, 2% effort

UM1AI069423-15S1 NIH/NIAID (Eron/Wohl/Herce)

02/01/2007 – 11/30/2027

HIV/AIDS Clinical Trials Unit (CTU) COVID-19 RadX-UP - Supplement

The major goals of this supplement are to improve testing access for underserved communities being missed by current testing approaches, and will generate new insights into a promising implementation strategy that can be leveraged to enhance ongoing pandemic response efforts and future delivery of COVID-19 vaccines and other prevention interventions once available.

Role: Co-investigator, 2% effort

2P30AI050410-24 NIH/NIAID (Swanstrom/Adimora)

08/20/2001 – 05/31/2026

The University of North Carolina Center for AIDS Research

The UNC Center for AIDS Research provides essential infrastructure support for the wide range of HIV related research for its members. This research contributes to the major goals of Ending the HIV Epidemic and HIV prevention and cure, and has overarching public health impact.

Role: Co-investigator, 2% effort

NIH/NIAID K01AI125087 (Edwards)

12/15/2016 – 12/14/2021 \$658,028 (direct)

Comparative effectiveness of tailored HIV treatment plans and mortality

The overall goal of this K01 application is to optimize clinical care decisions for people living with HIV. Specifically, this project will explore how antiretroviral therapy regimens affect cause-specific mortality and how treatment plans can be tailored or personalized based on patient characteristics to improve survival.

Role: Principal investigator, 60 - 85% effort

NIH UG3HL152189-01 (Mbaya)

07/01/20-06/30/22 \$403,104

Enhancing Blood Donation at Schools and Beyond: An Implementation Science Study

Inadequate blood supplies in Malawi continue to prevent optimal reduction of infant, under-5, maternal mortality and HIV associated deaths. We propose to conduct critical epidemiology surveillance and hemovigilance and qualitative research to understand facilitators and motivators of blood donation and strategies for engaging youth. Using this information, we will conduct a cluster randomized clinical trial to evaluate the effectiveness of an enhanced school-based donor program to increase both initial donation and long term retention as a donor.

Role: Co-Investigator, 5%

NC State Legislature (Edwards/Westreich)

6/1/2020 – 12/31/2020 \$597,641 (direct)

The Gillings Epidemiology Dashboard for NC Policymakers

The overall goal of this project is to build develop a resource center for SARS-CoV-2 testing, screening, and surveillance that will implement modern epidemiologic tools to enable better science and public health around SARS-CoV-2

Role: MPI, 2% effort

DHHS FDA 75F40119C10115 (Jonsson Funk)

09/23/19-03/22/21 \$1,275,489 Methodological Advances in the Assessment of Uncontrolled Confounding To address these needs and inform the FDA's upcoming (2021) draft guidance for industry regarding "the appropriate standards and methodologies for collection and analysis of real world evidence", we propose to evaluate and inform the use of sensitivity analyses used to assess the risk of uncontrolled confounding in observational studies.

Role: Co-investigator, 7% effort

NIH/NIAID CFAR Administrative Supplement (Edwards)

8/1/2019 – 7/31/2020 \$100,000 (direct)

Developing novel tools to understand the HIV Epidemic in the United States using Big Data

This project develops and illustrates tools to leverage novel fusions of detailed, rapidly updated HIV surveillance and cohort data. The project goals are to assess and track evolving demographics of HIV diagnoses and disparities in HIV treatment outcomes in the United States.

Role: Supplement PI, 5% effort

Gates Foundation PHSEZE81 (Weir)

01/01/15 – 06/30/17 \$2,133,021

MESH Consortium: Key Populations Working Group

The unique contribution of the consortium will be to maximize the potential of data routinely collected through HIV surveillance and service delivery platforms to provide timely information on key outcomes in relevant populations at a level appropriate to support resource allocation and realignment.

Role: Co-investigator, 14% effort

USAID AID-OAA-L-14-00004 (Thomas)

Monitoring and Evaluation to Assess and Use Results (MEASURE) Evaluation Phase IV Project Served as PI on 3 projects funded through the MEASURE Evaluation funding mechanism, 18% effort

Effect of mobility on treatment outcomes among mobile populations infected with HIV and TB in East Africa Cross-Border Regions

01/01/2018 – 3/31/2020 \$1,000,000 (total)

To estimate the impact of mobility on TB and HIV treatment outcomes among mobile and vulnerable populations coinfecting with TB and HIV in East Africa cross border regions.

Role: PI of project

Impact evaluation of the Cross Border Health Integrated Partnership Project

06/01/2015 – 12/31/2016 \$1,871,000 (total)

To estimate the impact of the cross border health integrated partnership project, a multifaceted structural intervention implemented in 10 cross border sites in 5 countries in East Africa.

Role: PI of project

Phylogenetic analysis of HIV infections in cross border sites in East Africa

08/01/2016 – 05/01/2017 \$790,000 (total)

To identify and describe HIV transmission clusters occurring among key populations in cross border sites in East Africa and to describe patterns of resistance to antiretroviral medications in these populations

Role: PI of project

Linkages across the continuum of HIV services for key populations affected by HIV (Weir)

Improving estimates related to the HIV care continuum in the Dominican Republic

10/01/2015 – 12/31/2016 \$300,000 (total)

To improve estimates of virologic suppression for sex workers and men who have sex with men in care for HIV in the Dominican Republic

Role: Activity Lead, 13% effort

UNC CFAR Developmental Award (Edwards)

11/1/2015 – 10/31/2016 \$30,000 (direct)

Examining causes of death among patients with HIV in the United States

To present critical information on trends in causes of death among patients with HIV in the United States.

Role: Principal investigator (no effort allowed)

New Aid Foundation Grant for the study of neglected tropical diseases (Edwards/Feinstein)

7/1/2010 - 7/1/2011 \$4000 (direct)

t. cruzi infection and cognitive development in primary school children in rural Nicaragua To provide information to local public health leaders on the prevalence of Chagas disease among children in rural Nicaragua and to assess the extent to which Chagas disease affects cognitive development.

Role: MPI

SERVICE

ACADEMIC SERVICE

Department of Epidemiology

Director of Graduate Studies, 2024 – present

Graduate Studies Committee, 2021 – present

Writing Committee, Methods Qualifying Exam, UNC Epidemiology, 2024

Writing Committee, Infectious Diseases Qualifying Exam, UNC Epidemiology, 2022

SARS-CoV-2 Return to Campus Modeling Group, 2020 - 2022

Sidney Kark Distinguished Teaching Assistant Award Committee, 2015

External dissertation reader, Yale School of Public Health, 2021 and 2024

Gillings School of Global Public Health

Doctoral Council, 2025 – present

Academic Programs Committee, 2024 – 2025

Tar Heel Tracker Degree Clearance Working Group, 2025 – present

Center for AIDS Research

Clinical Core Internal Advisory Board, 2022 - 2024

CFAR Developmental Award Review Committee, 2023 - 2024

EDITORIAL RESPONSIBILITIES

Associate editor, Epidemiology, 2022 - present

Associate editor, Frontiers in Epidemiology, Infectious Diseases, 2021 - present

Associate editor, American Journal of Epidemiology, 2020 - present

Associate editor, BMC Infectious Diseases, 2018 - 2022

Statistical advisor, PLoS Medicine, 2019 - present

Reviewer for American Journal of Epidemiology, Epidemiology, Biometrics, Statistics in Medicine, Annals of Applied Statistics, Journal of Applied Statistics, Journal of Causal Inference, Epidemiologic Methods, Journal of Acquired Immune Deficiency Syndromes Pharmacoeconomics and Drug Safety, Journal of Clinical Epidemiology, Annals of Epidemiology, Occupational and Environmental Medicine, Antiviral Therapy, Computational and Mathematical Methods in Medicine Lifetime Data Analysis, PLOS One, Scientific Reports, Environmental Health Perspectives, Paediatric and Perinatal Epidemiology, PLoS Medicine, Brazilian Journal of Infectious Diseases, and others

PROFESSIONAL SERVICE

Consultations and technical assistance

1. Consultation and technical support for WHO guidelines panel on infant postnatal prophylaxis for prevention of HIV. Dec. 2024 - April 2025
2. Key Population Surveillance and Estimates: Recent Advances and Future Directions Technical Working Group. CDC, Atlanta, GA, 24 May 2021
3. Co-investigator, Chatham County Community Cohort Study, Chatham County, NC, 2018 - present
4. UNAIDS Reference Group Meeting on Key Populations, Atlanta, GA, June 2, 2018
5. UNAIDS Reference Group Meeting on Estimates, Modeling and Projections, London, United Kingdom, 19 October 2017
6. Population Size Estimation Technical Consultation, CDC, Atlanta, GA, 31 January 2017
7. Strategic information for key populations technical consultation, UNAIDS, Geneva, Switzerland, 9 March 2016

Society for Epidemiologic Research

Executive Committee, Member-at-large & Liaison to Education Committee 2021 – 2024

Education committee co-chair 2021 – 2024

Publication committee member 2015 – 2018

Symposium session chair at annual meeting:

2025 - "Fake It 'Til You Make It: Making Progress when Data Have Gaps" 2024 - "EMERGE: Enhancing Measurement Error Research for Greater Epidemiologic Insight"

2023 - "Transforming education in a constantly evolving field: A multi-school perspective on how epidemiological curricula evolve and diversify over time in North America".

2022 - "On teaching advanced epidemiologic methods".

2019 - "The elephant in the room: Causal inference in the face of competing events"
2018 - "Epidemiologic research with incomplete and imperfect data: making progress in the face of uncertainty"
2017 - "Putting the 'Implementation' back into Implementation Science: Moving beyond the 'Implications' of Epidemiologic Research"
2016 - "Meeting in the middle: systems science and causal inference"
2015 - "Bayesian methods and causal inference"
2014 - "Selection bias due to loss: An old and often ignored problem revisited"
2013 - "Recording nature's answers: Measurement bias in epidemiology"
Oral abstract session organizer an annual meeting:
2025 - "Infectious Diseases"
Symposium session chair at mid-year meeting:
2022 - "Methods to address the dual pandemics of COVID-19 and health disparities"